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Palaeomagnetism of the basalts of Wadi Abu Tereifiya, Mandisha and dioritic dykes of Wadi Abu Shihat, Egypt

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Summary. One hundred and fifty oriented samples were collected from 12 sites from the Tertiary basalts of Wadi Abu Tereifiya (30.0° N, 32.1° E). After alternating field demagnetization the mean direction of the natural remanent magnetization is, $D = 187.9°$, $I = -20.8°$ with $\alpha_{95} = 5.8°$. This yields a palaeopole at 69.4° N, 188.3° E.

Also, 30 oriented samples were collected from two sites from Mandisha in Bahariya Oasis (28.4° N, 28.9° E). After cleaning, the mean direction of the NRM is $D = 191.0°$, $I = 5.2°$ with $\alpha_{95} = 9.9°$. This yields a palaeopole position at 58.2° N, 186.7° E.

Besides, the NRM of 70 oriented samples collected from seven dioritic dykes from Wadi Abu Shihat (26.3° N, 33.2° E) was found to have a mean direction, $D = 142.0°$, $I = -0.3°$, which leads to a palaeopole position at, 44.9° N, 273.0° E. This agrees with other Mesozoic pole positions from Africa.

1 Introduction

The present work is a continuation of the work done by the authors (Gouda Hussain *et al.* 1976a, b; Schult, Soffel & Gouda Hussain 1978; Schult 1974; Schult & Soffel 1973) and by El Shazli & Krs (1970, 1972) and Refai & Wassif (1972) to study the palaeomagnetism of the rocks in North Africa.

For the present work, oriented samples were collected from three areas in Egypt. They were:

(a) Wadi Abu Tereifiya (30° 00′ N, 32° 05′ E) NE Egypt near the Gulf of Suez. The area contains about 40 basaltic occurrences distributed between two faults forming a graben extending nearly NE–SW. The age of the basalt is thought to be from the Middle Eocene (El Sharkawi & Abu Kadra 1968).

(b) Mandisha (28° 22′ N, 28° 55′ E) in Bahariya Oasis, Western Desert. The basalt is in the form of a sheet overlying the Bahariya formation. Attia (1955) and El Akkad & Issawi

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(1964) showed that the basalt is post upper Eocene. The palaeomagnetism of some sedimentary rock units from the upper Eocene in Bahariya was measured in a previous work (Gouda Hussain 1977).

(c) The dioritic dykes of Wadi Abu Shihat (26° 20′ N, 33° 10′ E) Qena-Safaga District, Eastern Desert. The dykes are parallel, extending in a NS direction and cut the basement granites in the area. The geological age of the dykes is unknown.

Fig. 1 illustrates the locations of the three areas in Egypt and the sampling positions.

2 Sampling and measurements

The samples were collected using a drill and cores (2.5 cm diameter) were cut into specimens of 2.5 cm length. Some samples were collected as oriented hand samples and cut into cubes of 2.5 cm length. The magnetization was measured using a spinner magnetometer.

Some pilot samples were subjected to alternating field demagnetization in steps up to 1200 Oe and the magnetization was measured after each step. An appropriate field was chosen and all other specimens were demagnetized at this field to derive the stable remanent magnetization. The instruments and method of measurements are described by different workers, e.g. Collinson, Creer & Runcorn (1967) and Angenheister & Soffel (1972).

The magnetization of the basalts was found to be partially due to viscous magnetization and a typical example of the removal of this viscous magnetization by AF demagnetization

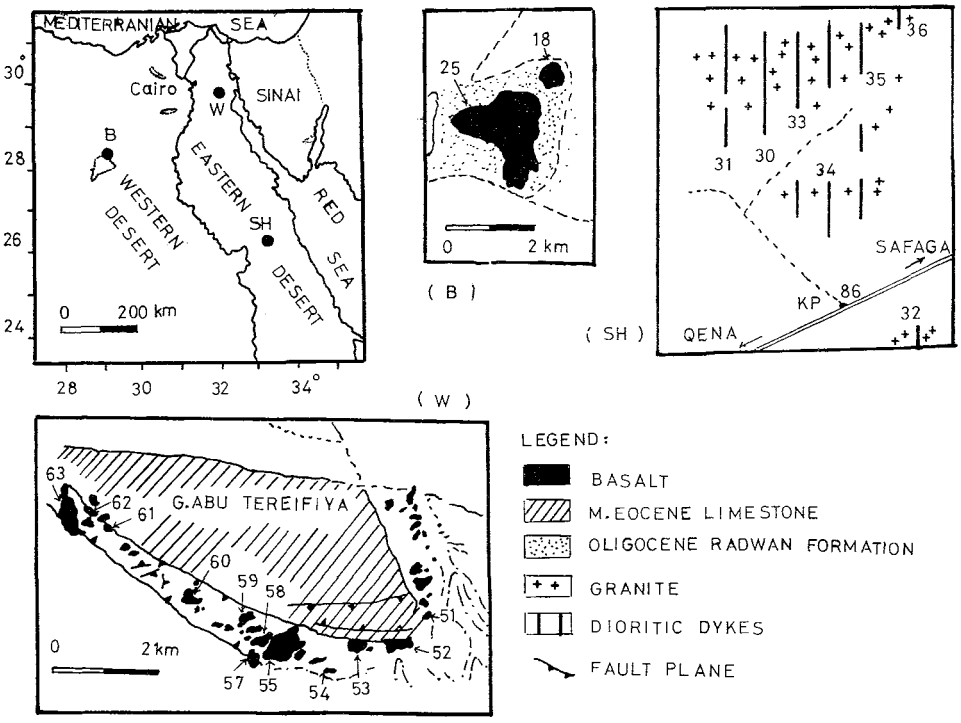


Figure 1. Map showing the locations of Wadi Abu Tereifiya (W), Mandisha, Bahariya (B) and Abu Shihat (SH), and the schematic geology maps of each area. Numbers refer to the sites where samples were collected.

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is given in Fig. 2(a). The relationship between the intensity of magnetization and the demagnetizing field is represented in Fig. 2(b). Fig. 3 illustrates stereographic projections of some directions of magnetization before demagnetization and the shift towards the directions of stable NRM.

The mean directions of magnetization after AF demagnetization for Abu Tereifiya (sites 51 to 63), Mandisha (sites 18 and 25) and Abu Shihat (sites 30 to 35) are represented by the stereographic projections in Fig. 4.

Table 1 contains the site means of palaeomagnetic results of the basalts and dioritic dykes under study.

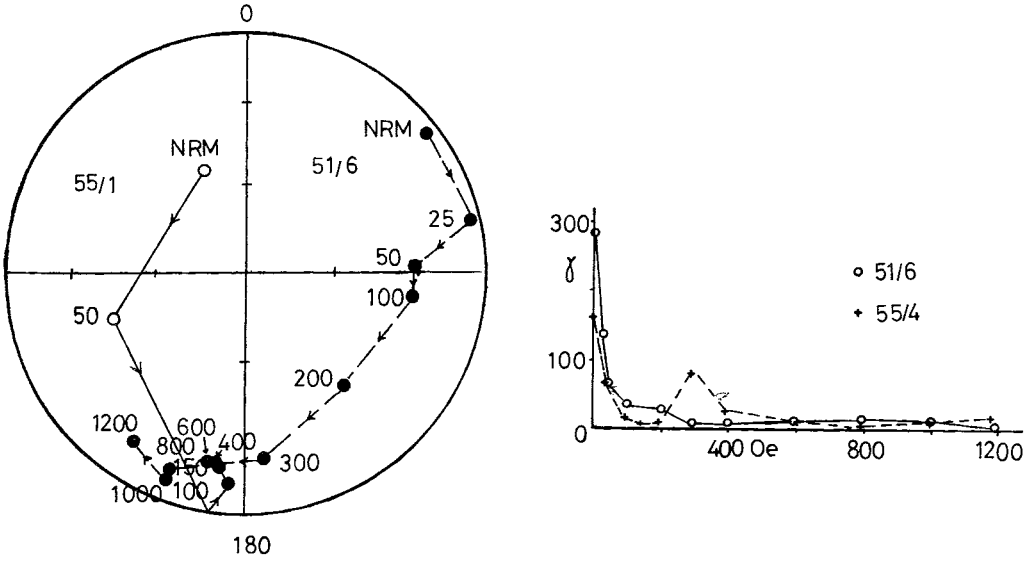


Figure 2. Stereographic projections of the directions of magnetization during alternating field (AF) demagnetization, and the intensity of magnetization as a function of demagnetizing AF.

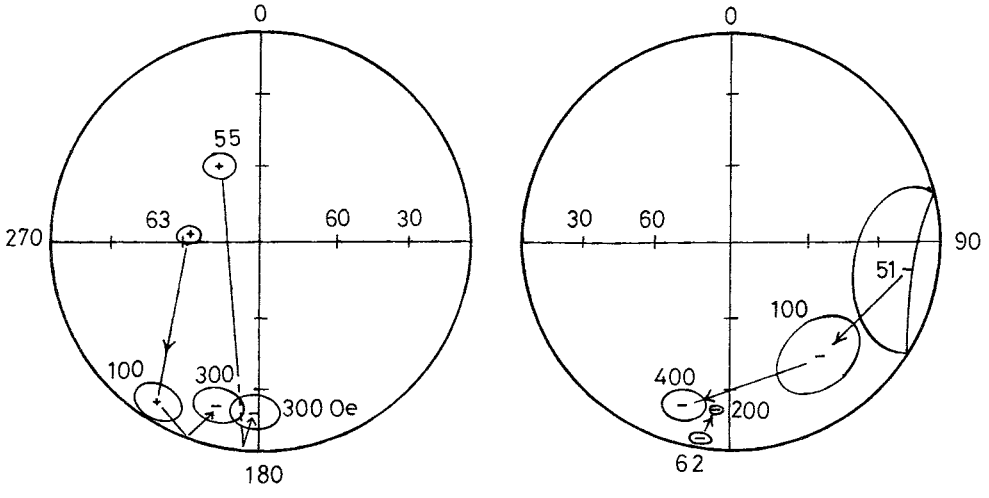


Figure 3. Stereographic projections of the directions of magnetization (site means and confidence circles) of the NRM and their shift during AF demagnetization.

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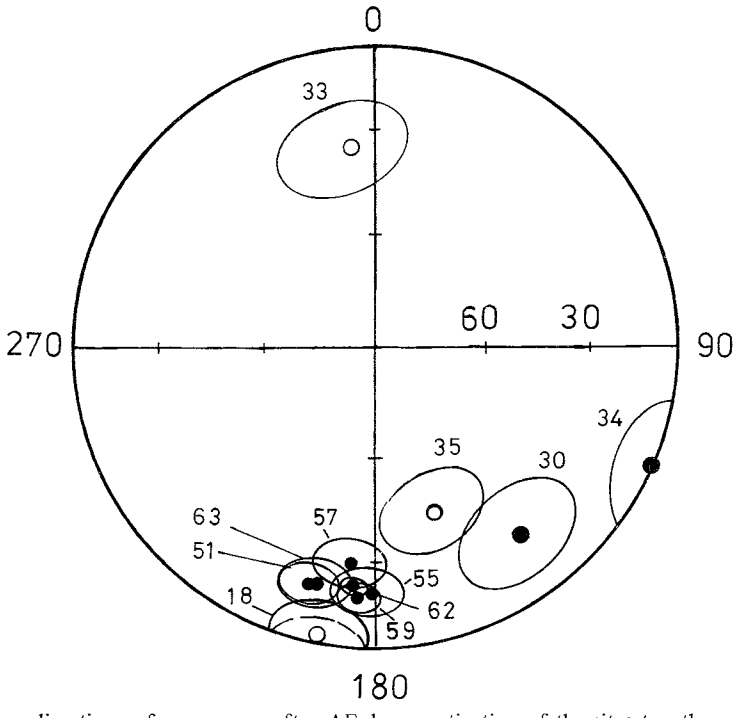


Figure 4. Mean directions of remanence after AF demagnetization of the sites together with the cones of confidence. Closed symbols: negative inclination, open symbols: positive inclination. Numbers refer to site numbers.

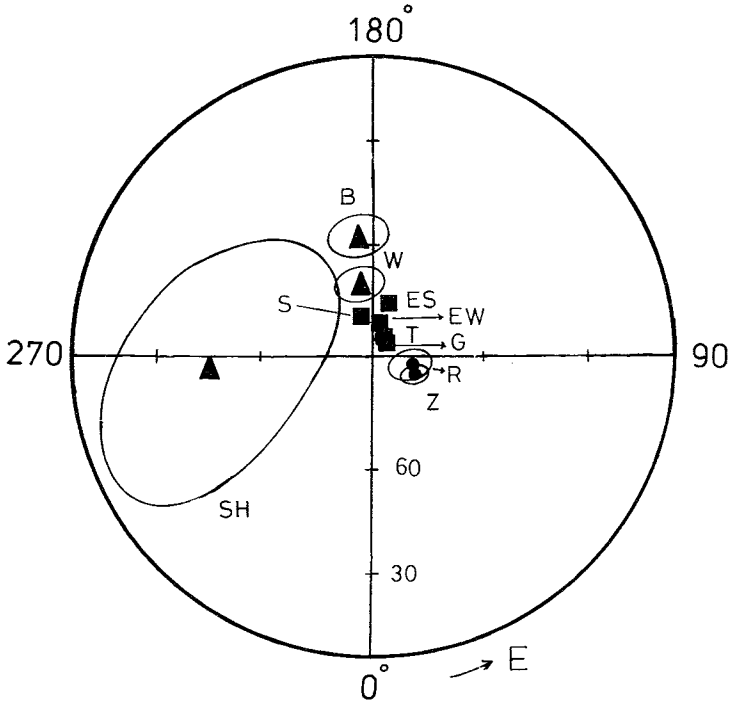


Figure 5. Means of palaeopole positions with their respective circles of confidence for Wadi Abu Tereifiya (W), Bahariya (B) and Shihat (SH). The pole positions for Ethiopian East and West (ES and EW), Turkana (T), Garian (G) Lybia, Rawash (R) and Zaabal (Z) Egypt are given for comparison.

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3 Discussion

The basalts have rather unstable magnetization due to a strong viscous magnetization. Some negative results were obtained for occurrences from Abu Tereifiya and Abu Shihat areas. All the significant results are tabulated in Table 1 and represented in Figs 2 to 4.

The mean palaeopole positions calculated for each locality are represented together with their circles of confidence in Fig. 5. These palaeomagnetic positions are compared with the other pole positions in Africa, e.g. ES: Ethiopian South, EW: Ethiopian West, T: Turkana, G: Garian, Lybia (Schult & Soffel 1973), A: Aswan, R: Rawash and Z: Zaabal, Egypt (Gouda Hussain *et al.* 1976b). The pole wandering curve of Africa represented in Fig. 6 (McElhinny 1973) indicates a Tertiary age for the basalt of Abu Tereifiya and Mandisha, and a Mesozoic age of the dioritic dykes of Wadi Abu Shihat.

Table 1. Site mean palaeomagnetic results of the basalts of Wadi Abu Tereifiya, Mandisha and Abu Shihat.

Site	Location		N	NRM				α_{95}	Pole position	
	°N	°E		Cleaning AF Oe	J_r 10^{-5} gauss	D(°)	I(°)	(°)	°N	°E
51	30.0	32.1	12	400	45.9	196.0	−18.8	7.6	64.8	172.4
52	30.0	32.1	7	100	4.1	no consistent result				
53	30.0	32.1	5	100	135.2	no consistent result				
54	30.0	32.1	12	100	361.5	no consistent result				
55	30.0	32.1	9	270	32.5	182.1	−19.3	8.3	69.9	206.0
57	30.0	32.1	6	150	94.5	186.9	−29.1	8.3	74.3	186.8
58	30.0	32.1	20	150	80.0	no consistent result				
59	30.0	32.1	9	200	108.0	183.1	−16.9	5.2	68.5	203.7
60	30.0	32.1	25	100	57.3	no consistent result				
61	30.0	32.1	5	50	48.6	no consistent result				
62	30.0	32.1	54	200	179.3	185.3	−20.2	2.5	62.9	196.8
63	30.0	32.1	21	300	19.7	194.1	−19.9	8.9	66.4	175.4
Means						187.9	−20.8	5.8	69.4	189.4
						$A_{95} = 4.6, d_p = 3.2, d_m = 6.1$				
18+25	28.3	28.9	730	150		191.0	5.2	9.9	58.2	186.7
30	26.3	33.2	7	160	9.9	142.7	−20.9	15.5	51.6	286.5
31	26.3	33.2	3	100	3.0	336.0	18.4	35.0	61.7	270.9
32	26.3	33.2	15	100	6.1	no consistent result				
33	26.3	33.2	15	100	12.4	no consistent result				
34	26.3	33.2	10	200	119.5	113.3	−0.6	13.2	20.8	292.0
35	26.3	33.2	10	150	30.9	160.0	40.5	12.9	36.8	236.0
36	26.3	33.2	8			no consistent result				
Means	(sites 30–36)					142.0	−0.3	43	44.9	273.0
						$A_{95} = 32.3, d_p = 21.5, d_m = 43.0$				

V: number of samples, J_r: remanent magnetization, D: declination, I: inclination, α_{95} or A_{95}: radius of the 95 per cent circle of confidence, NRM: natural remanent magnetization after alternating field (AF) cleaning.

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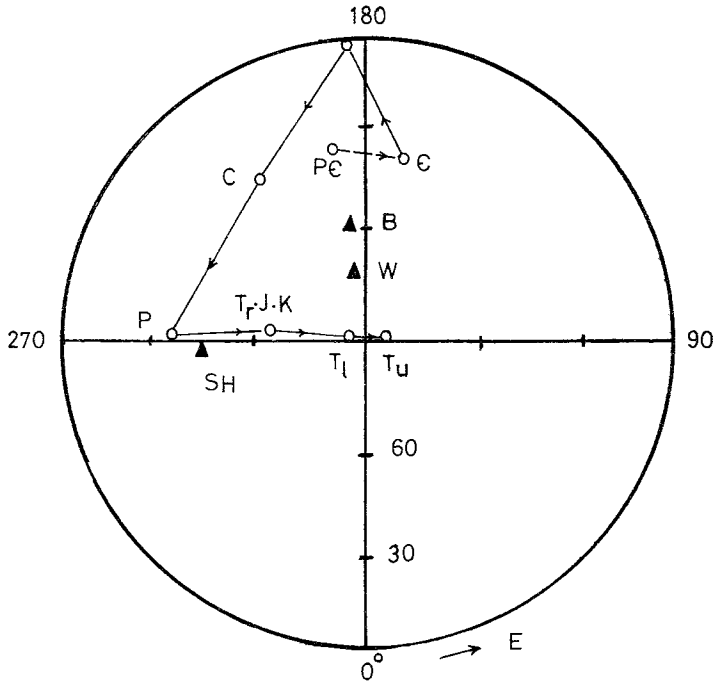


Figure 6. Polar wander path for Africa and the pole positions of the rocks under investigation.

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